Unnamed_Unit

Unit 3 - Creating Color-Changing LED Lights

PDF Documents

PDF Document 1: 1738217957210-Chapter 5 - Unit 3 - Creating Color-Changing LED Lights.pdf

This PDF document is included in the unit materials.

UNIT 3 CREATING COLOR-CHANGING LED LIGHTS

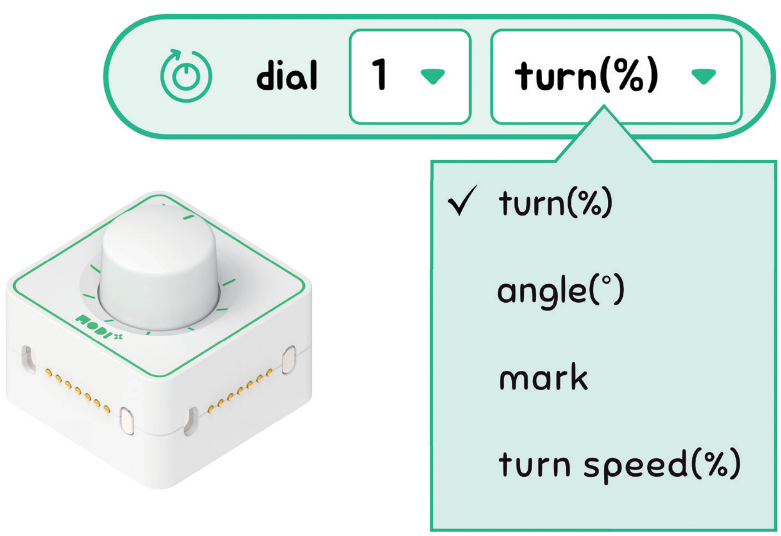
LED lights can produce various colors, making them practical tools for setting the mood. In this project, we' ll build an LED lighting system that allows you to create and adjust colors to your liking.

HOW DOES IT WORK

This project aims to adjust the colors to create lighting that suits the desired atmosphere. We' ll use a dialog sensor to control the colors of the LED module. Depending on the angle of the dialog sensor, the color will change, and a button will be added to turn the lights on and off.

Module Functionality

Let's look detail about the Dial module.

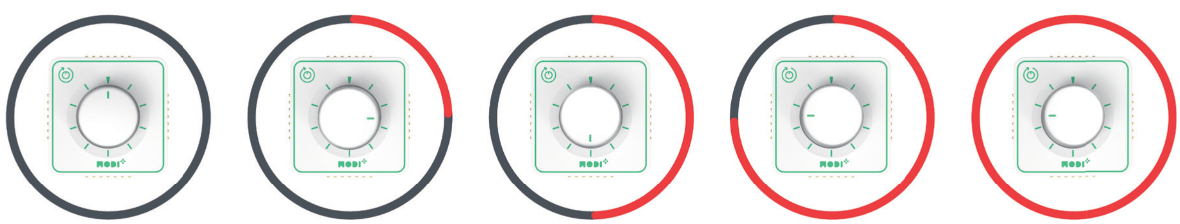


FUNCTIONS IN DETAIL

1. Turn (%) - Range: 0% ~ 100%	**2. Angle (°) - Range**: 0~360°
This just means moving the dial around, either to the right or left.	This is how far you've moved the dial from the default (0°), measured in degrees.

3. Mark - Range: 1~10	**4. Turn Speed (%) - Range**: 0% ~ 100%
Marks on the dial show where it can be turned to, like the numbers on a clock.	This tells us how fast you are turning the dial, in percentage.

TURN (%)



0% 25% 50% 75% 100%

ANGLE (°)

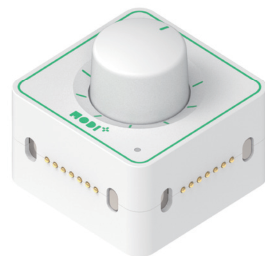
0° 90° 180° 270° 360°

The Dial module can rotate from 0 to 360 degrees, similar to turning a knob full circle. For simplicity, we'll consider a full rotation as 100%. Therefore, a half-turn, or 180 degrees, represents 50% of a full rotation.

Material Needed


Network


Button


Dial

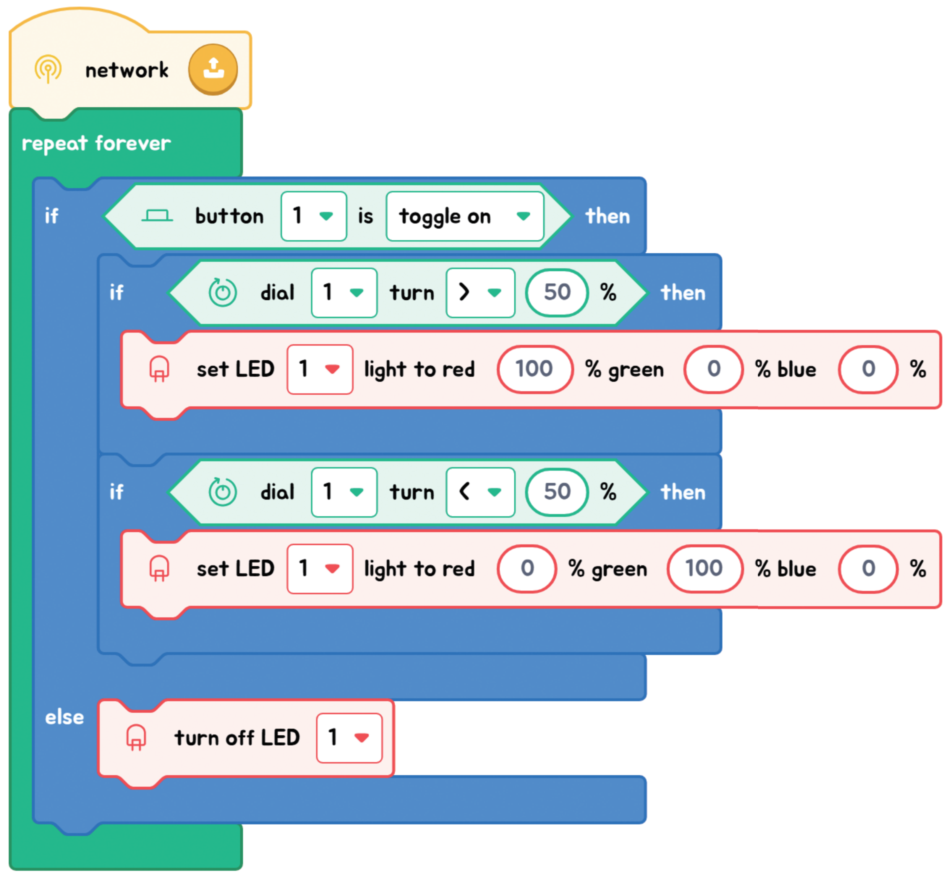

LED

When we combine all the modules, a creation has to look like this.

Code Story

If the toggle button is on	
↓ **True**	↓ **False**
Led ON	
Dialog < 50%	
Red	
	Led OFF
Dialog > = 50%	
Green	

Code Quest



Checkpoint

1. Extend the LED lighting system to include four additional colors. Modify the current dialog conditions to a "Mark-Range" setup for easy color selection based on specific ranges.

2. Create an automatic lighting system using a Time-of-Flight (ToF) sensor. The light should turn on when a person passes by and automatically turn off after five seconds.